Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Data Storytelling with Tableau

TGS Ref No: TGS-2020505550

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
17.0	15 August 2026	Rebuilt from legacy v16 with current visual deck, ten Excel-backed Tableau labs and aligned WA/PP assessment.	Dr Alfred Ang

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Course Overview

Data Storytelling with Tableau is a two-day, 16-hour WSQ programme aligned to Data Storytelling (ICT-DIT-4006-1.1). Learners progress from story framing and visual insight to dashboard development, stakeholder review and an interactive Tableau capstone using realistic synthetic Excel datasets.

Course Code	TGS-2020505550
Training Duration	16 hours across 2 days
Assessment	WA 60 minutes; PP 90 minutes
Trainer	Dr Alfred Ang
Mode	Instructor-led classroom with individual Tableau labs using supplied Excel sources

Learning Outcomes

- LO1: Utilise Tableau for data visualisation.
- LO2: Identify trends and patterns from data.
- LO3: Develop dashboards and stories for data storytelling.
- LO4: Present stories to stakeholders and review them with subject-matter experts.
- LO5: Design interactive visualisations for data storytelling.

Daily Schedule

Day 1: Story strategy, Tableau foundations, visual insight and dashboard design

Time	Duration	Topic / Activity	Method	Slides
9:00-9:20	20 min	Welcome, AM digital attendance, introductions and outcomes	Facilitated briefing	—
9:20-10:30	70 min	Topic 1: story purpose, audience, narrative and Tableau workflow	Concept briefing, worked examples and questioning	Slides 18-39
10:30-10:40	10 min	Tea break	Break	—
10:40-12:30	110 min	Labs 1-2: retail and campaign story framing	Demonstration, individual practice and debrief	Slides 35-38
12:30-13:15	45 min	Lunch break	Lunch	—
13:15-14:20	65 min	Topic 2: visual perception, chart selection and insight patterns	Concept briefing, worked examples and questioning	Slides 40-61
14:20-15:10	50 min	Labs 3-4: service trends and Singapore delivery map	Demonstration, individual practice and debrief	Slides 57-60
15:10-15:20	10 min	Tea break	Break	—
15:20-16:20	60 min	Topic 3: dashboard process, layout, filters and story points	Concept briefing, worked examples and questioning	Slides 62-83
16:20-17:45	85 min	Lab 5: omnichannel executive dashboard	Demonstration, individual practice and debrief	Slides 79-80
17:45-18:00	15 min	Day 1 reflection, Q&A and PM digital attendance	Reflection and Q&A	—

Scheduled learning and assessment time excluding lunch: 495 minutes.

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Day 2: Narrative review, calculations, parameters, actions and interactive capstone

Time	Duration	Topic / Activity	Method	Slides
9:00-9:15	15 min	Day 1 recap and AM digital attendance	Facilitated briefing	—
9:15-10:20	65 min	Topic 3 continued: dashboard actions, tooltips, stories and quality	Concept briefing, worked examples and questioning	Slides 62-83
10:20-10:30	10 min	Tea break	Break	—
10:30-11:20	50 min	Lab 6: branch turnaround Tableau Story	Demonstration, individual practice and debrief	Slides 81-82
11:20-12:30	70 min	Topic 4: presentation, stakeholder questions and SME review	Concept briefing, worked examples and questioning	Slides 85-106
12:30-13:15	45 min	Lunch break	Lunch	—
13:15-14:05	50 min	Labs 7-8: hospital presentation and SME evidence review	Demonstration, individual practice and debrief	Slides 102-105
14:05-14:50	45 min	Topic 5: calculations, parameters, actions and interactive UX	Concept briefing, worked examples and questioning	Slides 107-128
14:50-15:20	30 min	Labs 9-10: pricing explorer and regional capstone briefing	Demonstration, individual practice and debrief	Slides 124-127
15:20-15:30	**10 min**	**Assessment briefing and digital attendance**	**Individual summative assessment**	**—**
15:30-16:30	**60 min**	**Written Assessment (WA/SAQ), open book**	**Individual summative assessment**	**—**
16:30-18:00	**90 min**	**Practical Performance (PP), open book**	**Individual summative assessment**	**—**

Scheduled learning and assessment time excluding lunch: 495 minutes.

Topic-by-Topic Delivery Guide

Topic 01: Data Storytelling Using Tableau · Slides 18-39

Story purpose · audience · evidence · storyboard · Tableau workflow

- **Story purpose:** Connect a decision, evidence, explanation and call to action.
- **Audience:** Match depth, vocabulary and interaction to stakeholder needs and data literacy.
- **Story arc:** Orient the audience, reveal the change, explain the driver and land the decision.
- **Storyboard:** Sketch the sequence before building worksheets and dashboards.
- **Tableau workflow:** Connect, explore, build views, assemble dashboards, create story points and share.
- **Tool fit:** Choose Tableau when governed visual exploration and interactive communication are needed.

Lab 1: Frame a Retail Performance Story · Slides 35-36

Alignment: LO1 · A1 · K1. Learner output: A validated Tableau data connection, audience brief, story objective and three-point storyboard. Acceptance: A reviewer can state the audience, decision, evidence grain and three-point narrative without opening Tableau.

Lab 2: Build a Campaign Insight Narrative · Slides 37-38

Alignment: LO1 · A1 · K1. Learner output: Three worksheets and a story outline linking reach, conversion and profitable customer segments. Acceptance: The narrative identifies one channel to scale, one audience to protect and one risk or limitation to monitor.

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Topic 02: Drawing Insights from Visualisation · Slides 40-61

Chart selection · perception · patterns · trends · comparisons · uncertainty

- **Question first:** Select a chart from the analytical question, not from visual novelty.
- **Comparison:** Use position and length to make category differences easy to judge.
- **Trend:** Use time on a continuous axis and preserve meaningful baselines and intervals.
- **Relationship:** Use scatterplots to inspect association, clusters and unusual observations.
- **Composition:** Show part-to-whole only when the total and categories are stable and meaningful.
- **Annotation:** Direct attention to the evidence without hiding context or uncertainty.

Lab 3: Diagnose Service Trends and Outliers · Slides 57-58

Alignment: LO2 · A2 · K2. Learner output: A trend view, distribution view and annotated exception view with a defensible operational insight. Acceptance: The final view distinguishes trend, seasonality and exceptions and does not claim causation from correlation.

Lab 4: Map Delivery Performance Across Singapore · Slides 59-60

Alignment: LO2 · A2 · K2. Learner output: A proportional-symbol map, ranked companion chart and geographic insight with a stated data-quality caveat. Acceptance: The map and ranking agree on the highest-priority planning areas and the caveat prevents over-precise interpretation.

Topic 03: Develop Dashboards and Stories · Slides 62-83

Dashboard architecture · layout · context · story points · narrative sequence

- **Dashboard purpose:** One primary decision and a small set of supporting questions.
- **Information hierarchy:** Place the decisive message first, then explanation and detail.
- **Layout:** Use containers, alignment, whitespace and predictable reading order.
- **Context:** Expose period, scope, definitions, filter state and data freshness.
- **Story points:** Sequence views or dashboards so each point advances the argument.
- **Performance:** Limit marks, heavy filters and redundant views so interaction stays responsive.

Lab 5: Assemble an Omnichannel Executive Dashboard · Slides 79-80

Alignment: LO3 · A3 · K3. Learner output: A 1366 x 768 dashboard with KPI strip, revenue trend, channel comparison, store drill-down and visible context. Acceptance: The dashboard answers the executive question in under 30 seconds and exposes period, filters, definitions and freshness.

Lab 6: Create a Branch Turnaround Tableau Story · Slides 81-82

Alignment: LO3 · A3 · K3. Learner output: A four-point Tableau Story with captions, annotations, saved filter states and a clear call to action. Acceptance: Each story point advances the argument, the filter state is intentional, and the last point names an owner and next action.

Topic 04: Present and Review Stories · Slides 85-106

Stakeholder narrative · presentation · SME review · evidence traceability

- **Executive opening:** Lead with the decision, impact and confidence level.
- **Evidence ladder:** Move from KPI to trend, driver, segment and record-level detail.
- **Interaction discipline:** Use filters to answer likely questions without losing the narrative.
- **SME review:** Separate data facts, analytical interpretation and domain judgement.
- **Revision log:** Record what changed, why, who reviewed it and which evidence supports it.
- **Accessibility:** Use descriptive titles, sufficient contrast, limited marks and non-colour cues.

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Lab 7: Present a Hospital Flow Story · Slides 102-103

Alignment: LO4 · A4 · K4. Learner output: A stakeholder-ready dashboard, five-minute presentation script and question/evidence log. Acceptance: The presentation answers the operations question, identifies a limitation and records unresolved SME questions for follow-up.

Lab 8: Run an SME Review and Evidence Revision · Slides 104-105

Alignment: LO4 · A4 · A6 · K4. Learner output: A reviewed dashboard, metric-definition register, decision log and v2 release note. Acceptance: Every material claim has a definition, evidence source and reviewer; the change log distinguishes correction from interpretation.

Topic 05: Create Interactive Stories · Slides 107-128

Calculated fields · parameters · filters · actions · navigation · user experience

- **Calculated fields:** Create governed measures and business logic when source fields are insufficient.
- **Parameters:** Expose a controlled value that can drive calculations, filters or reference lines.
- **Filter actions:** Use a selected mark in one view to constrain another view.
- **Highlight actions:** Preserve context while emphasizing related marks.
- **URL and navigation:** Connect the dashboard to detail, definitions or another analytical view.
- **Experience design:** Make state, affordance, reset and next action visible to the user.

Lab 9: Build a Parameter-Driven Pricing Explorer · Slides 124-125

Alignment: LO5 · A5 · K5. Learner output: A dashboard with metric parameter, discount threshold, dynamic title, parameter action and reset guidance. Acceptance: A user can switch the metric, change the threshold, select a product and reset the view without editing a worksheet.

Lab 10: Deliver an Interactive Regional Story Capstone · Slides 126-127

Alignment: LO1–LO5 · A1–A6 · K1–K5. Learner output: A packaged Tableau workbook with executive dashboard, geographic drill-down, interactive story and signed review checklist. Acceptance: The packaged workbook opens with embedded data; interactions work; the story answers the decision; and A1–A6 evidence is traceable.

Resources Required

- Tableau Public or Tableau Desktop on Windows or macOS.
- Microsoft Excel or compatible software for inspecting each synthetic data source.
- Learner Guide and the ten numbered lab folders.
- An Excel workbook plus Tableau TWB and packaged TWBX sample in every lab folder.
- Projector for concept briefings, demonstrations and stakeholder presentations.
- LMS/TMS access for courseware, attendance and assessment submission.

Assessment

- Written Assessment (WA/SAQ) — 5 open-ended questions, 60 minutes, open book.
- Practical Performance (PP) — 5 connected Tableau tasks, 90 minutes, open book.
- Assessment starts at 3:30 PM on Day 2 and ends at 6:00 PM, preserving the approved 60-minute WA and 90-minute PP timings.
- Assessment is individual and open book. Learners submit completed papers through the LMS.
- A minimum of 75% attendance and a Competent assessment outcome are required.
- https://lms-tms.tertiaryinfotech.com/

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